

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Seventh Semester of B. Tech. (CL) Examination

May 2014

CL421 Advanced Structural Analysis

Date: 24.05.2014, Saturday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.
5. All required sketches/data are available at the end of the paper.

SECTION – I

Q - 1 Enumerate types of bridges and explain components of suspension bridge with neat sketch. [07]

Q - 2 Calculate design bending moment and design shear force for following data of deck slab. [14]

Clear span – 4 m

Width of footpath – 600 mm on either side

Wearing coat – 80 mm

Road Width = 7.5 m

Loading – IRC class AA (Tracked vehicle)

Material – Concrete grade M 25 & Fe 415 grade HYSD bars

Assume the thickness of slab to be 80 mm per meter span of the slab.

OR

Q - 2 For IRC Class A loading, using the same data of above Question 2, calculate design bending moment and design shear force. [14]

Q - 3 Attempt any Two: [14]

- (i) Explain analysis of slabs spanning in two directions.
- (ii) Explain Hendry Jaeger Method for the analysis of a tee beam and slab deck bridge
- (iii) Explain Corbon's Method for the analysis of a typical tee beam and slab deck bridge.

SECTION – II

Q - 4 What is non-linear analysis? Explain types of non – linearity. [07]

Q - 5 (a) Enlist the assumption made in non-linear analysis. [02]

(b) Enlist methods of solution of non-linear analysis and explain any one method in detail. [06]

- (c) Derive an expression for Meridional thrust and hoop force for a spherical dome [06]
subjected to uniformly distributed load at crown.

OR

- (c) A roof of a hall diameter 30 m is to be covered by a conical dome of 120 mm thickness [06]
and 5 m rise. The live load acting is 2.5 kN/m^2 . Calculate stresses in the dome.

Q – 6 Attempt any Two: [14]

- (i) Design a conical roof for a hall having diameter of 16 m and the rise of the dome is 3.5 m. Take the live load as 1800 N/m^2 . Use M 20 grade concrete and Fe 415 steel.
- (ii) Design a spherical dome for a hall of 12 m diameter. The rise of the dome is to be 1.6 m. The live load is 1000 N/m^2 and floor finishes as 200 N/m^2 . The dome carries a lantern load of 10 kN concentrated at the crown. Use M 20 grade concrete and Fe 415 steel.
- (iii) Derive an expression for Meridional thrust and hoop force for a conical dome subjected to uniformly distributed load at crown.

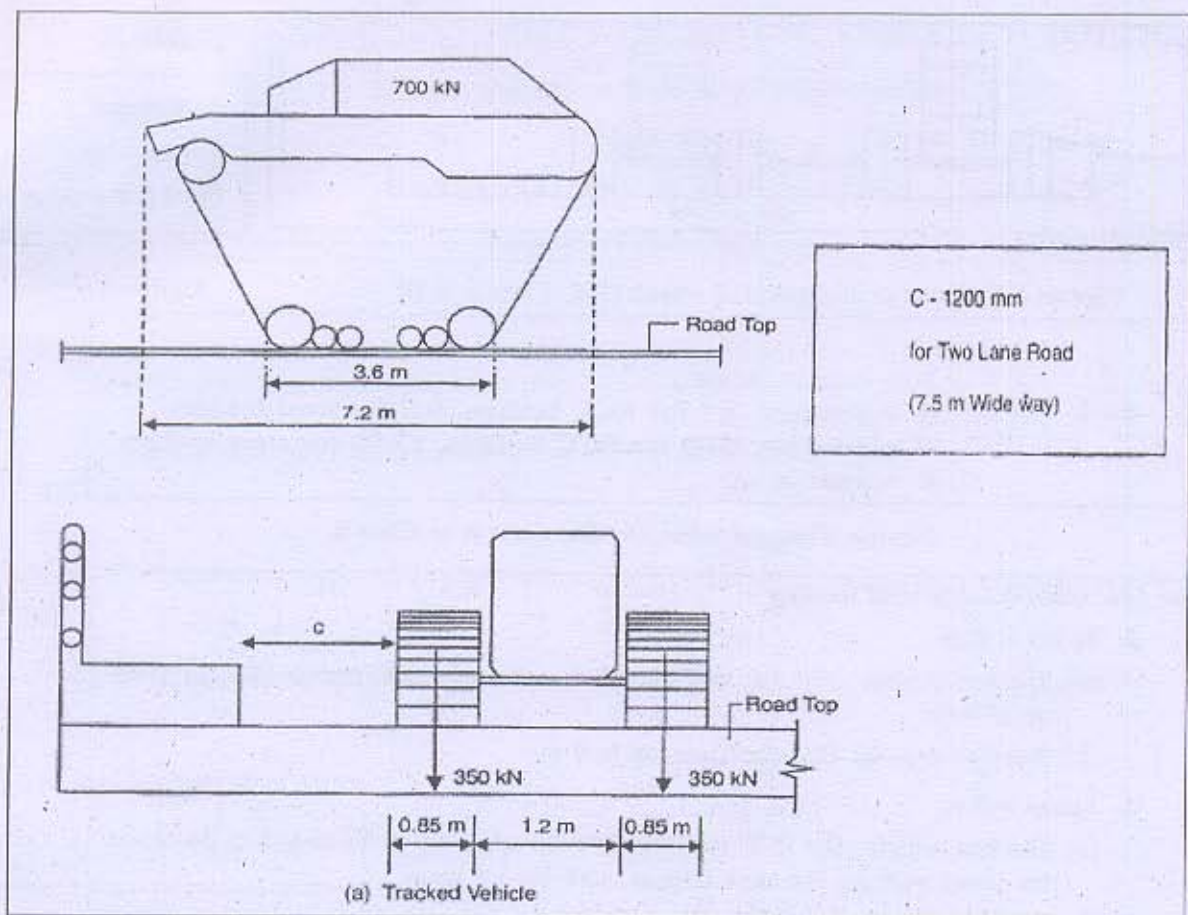


Figure: 1 Class AA (Tracked)

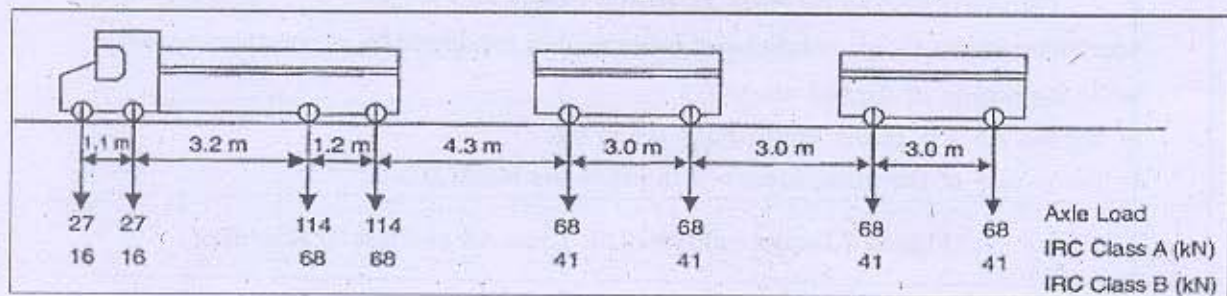


Figure: 2 Side View Details (IRC Class A & B)

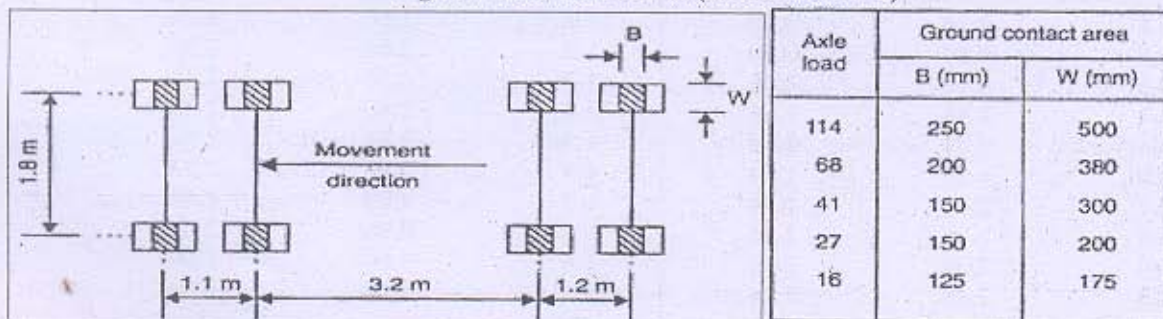


Figure: 3 Wheel impressions on road (IRC Class A & B)

Table: 1

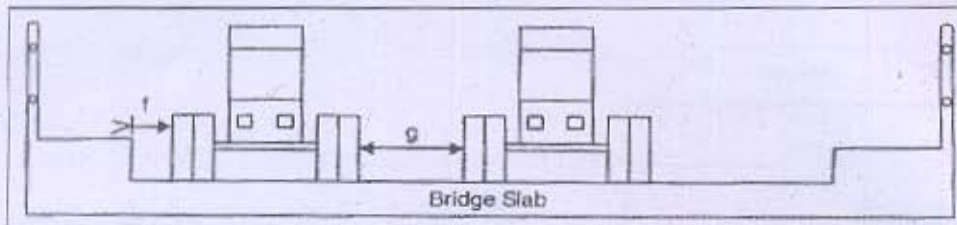


Figure: 4 Transverse placement of wheels (IRC Class A & B)

$f = 150 \text{ mm}$
 $g = 1200 \text{ mm}$
 (for 7.5 m wide way)

Figure: 5

where

$$I_f = \frac{A}{B + L}$$

I_f = impact factor
 A = constant, 4.5 for RCC bridges, 9.0 for steel bridges
 B = constant, 6.00 for RCC bridges, 13.50 for steel bridges
 L = span in m.

Figure: 6 Impact value for IRC Class A or Class B

For IRC Class AA and 70R loading

1. Spans < 9 m

- (a) *Tracked vehicle*. 25% for spans up to 5 m linearly reducing to 10% for spans up to 9 m.
- (b) *Wheeled vehicle*. 25% for spans up to 9 m.

2. Spans > 9 m

- (a) *Tracked vehicle*. For RCC bridges, 10% for spans up to 40 m and as per graph for spans > 40 m. For steel bridges, 10% for all spans.
- (b) *Wheeled vehicle*. For RCC bridges, 25% for spans up to 12 m, and in accordance with graph for spans > 12 m. For steel bridges, 25% for spans up to 23 m and as per graph (IRC 6) for spans exceeding 23 m.

Appropriate impact factors as mentioned below need be considered for substructures as well.

- > At the bottom of the bed block: 0.5
- > For the top 3 m of the substructure: 0.5 to 0.0
- > For portion of the substructure > 3 m below the block: 0.0

Figure: 7 Impact value for IRC Class AA or Class 70 R loading

B/l	α for sss*	α for cs*	B/l	α for sss	α for cs
0.1	0.40	0.40	1.1	2.60	2.28
0.2	0.80	0.80	1.2	2.64	2.36
0.3	1.16	1.16	1.3	2.72	2.40
0.4	1.48	1.44	1.4	2.80	2.48
0.5	1.72	1.68	1.5	2.84	2.48
0.6	1.96	1.84	1.6	2.88	2.52
0.7	2.12	1.96	1.7	2.92	2.56
0.8	2.24	2.08	1.8	2.96	2.60
0.9	2.36	2.16	1.9	3.00	2.60
1.0	2.48	2.24	2.0 and above	3.00	2.60

* sss = simply supported slab, cs = continuous slab

Table: 2 Values of α for slabs (IRC 21)